

Implementation of Antiferromagnetic Barrier with High Tunneling Magnetoresistance: A Pathway Toward Magnon-Assisted Spin-Transfer Torque Switching

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Low-energy switching of magnetic tunnel junctions (MTJs) remains pivotal for applications such as magnetoresistive random access memory (MRAM) and hybrid MTJ/CMOS systems. Although spin-transfer torque (STT) switching presents a promising method for writing operations, it is challenged by the high current density requirements. Magnon-assisted STT switching through the insulating barrier is a viable method due to high temperature gradient between the ferromagnetic electrodes. However, efficient magnon transfer requires an antiferromagnetic barrier. This study focuses on fabricating MTJs with antiferromagnetic barrier to realize magnon-assisted STT switching. At room temperature, our MTJs with NiO barrier exhibit a substantial 26% Tunneling Magnetoresistance (TMR), marking one of the highest reported values among MTJs employing NiO barriers.

Index Terms—Antiferromagnet, Magnetic Tunnel Junction, Tunneling Magnetoresistance, Magnon-assisted STT switching.

I. INTRODUCTION

ENERGY efficient switching of magnetic tunnel junctions (MTJs) remains pivotal for the advancement of applications such as magnetoresistive random access memory (MRAM) and hybrid MTJ/CMOS systems [1]. Although spin-transfer torque (STT) switching presents a promising method for memory writing operations, it is fundamentally challenged by the high current density requirements [2]. Using magnons to assist STT switching is an energy-efficient alternative, provided a viable method is established [3].

During standard STT switching, ballistic electron tunneling naturally creates a temperature differential between the fixed and free layers of the MTJ. Implementing a magnetic barrier allows for the transfer of magnons to the free layer, with the magnon flow moving in the opposite direction to the electron flow, as illustrated in Fig. 1 [3], [4]. To enable independent control of the magnetization directions in these layers, the barrier cannot be ferromagnetic; however, an antiferromagnetic barrier is highly suitable. To eliminate the adverse effects of exchange bias (H_{EB}) during standard operation, the blocking temperature (T_B) of the chosen AFM material must be lower than room temperature. Thin-film NiO fulfills these criteria perfectly as a room-temperature antiferromagnet (AFM).

II. RECORD HIGH TMR IN MTJs WITH NiO ANTIFERROMAGNETIC BARRIER

The most significant achievement of this study is the realization of a substantial 26% Tunneling Magnetoresistance (TMR) at room temperature, shown in Fig. 2. This is the highest reported TMR ratios among MTJs with only NiO tunneling barriers.

Samples were deposited via reactive-RF and DC magnetron sputtering on thermally oxidized Si wafers with the stack: Ta(8)/Ru(7)/Ta(9)/Co₂Fe₆B₂(5)/NiO(3)/Co₂Fe₆B₂(4)/Ta(4)/Ru(15) (thicknesses in nm). Reactive sputtering with

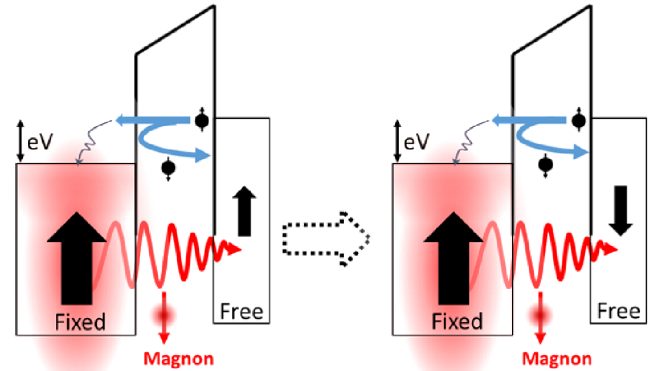


Fig. 1. The schematic of Magnon-assisted STT switching of the MTJ from parallel to antiparallel state. The fixed layer is hotter than the free layer due to the ballistic electron tunneling. Magnon flow is opposite to the electron flow (Schematic is adapted from [4]).

5% oxygen-to-argon ratio was required for a high quality barrier due to an oxygen-deficiency in the NiO barrier. After 375°C annealing, Co₂Fe₆B₂ (CFB) films with in-plane anisotropy exhibit a notably high coercive field (Fig. 2). The behavior is attributed to the antiferromagnet with low anisotropy energy due to the low thickness of the antiferromagnetic layer. Unpinned interfacial spins switch with the ferromagnet instead of inducing exchange bias, which enhances the coercivity. Both the room temperature and the low temperature hysteresis loops demonstrate a high coercive field compared to in-plane anisotropy thin films with no exchange bias. Further investigation of the antiferromagnetic phase and the quality of the barrier is analyzed by decreasing the anisotropy energies of the ferromagnetic layers. The as-deposited CFB layers have lower anisotropy energy since annealing increases the anisotropy energy of CFB by crystallization and interfacial orbital hybridization.

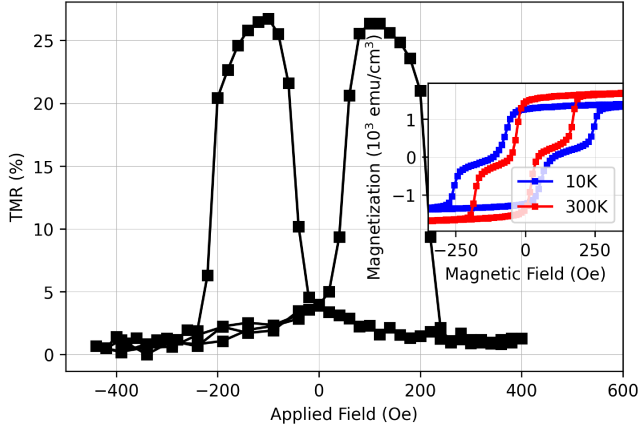


Fig. 2. High TMR and Magnetic Hysteresis: Room-temperature TMR curve demonstrating the 26% ratio, obtained from a 50 μm diameter MTJ. Inset: M vs H hysteresis loops from a full-film at room-temperature and at 10K and 300K.

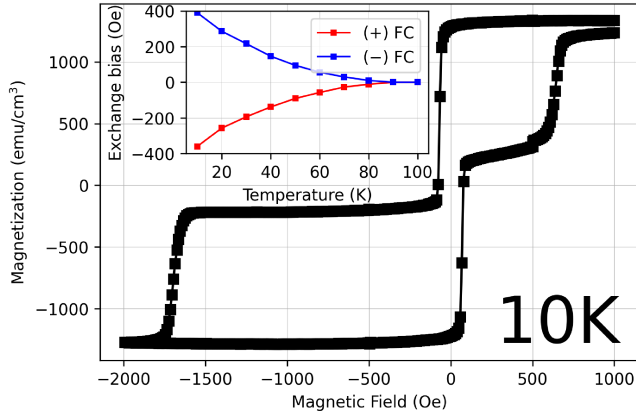


Fig. 3. Low-Temperature Hysteresis at 10 K for the as-deposited, full-film sample. The inset shows exchange bias versus temperature.

A. Exchange Bias and The Anisotropy Dependence

Hysteresis of as deposited full-film At 10 K shows two distinct switchings (Fig. 3): one CFB layer exhibits an exchange bias shift, while the other switches symmetrically. In standard MTJs, the bottom CFB has higher magnetic anisotropy due to the smooth underlying buffer layer. The subsequent deposition of the NiO barrier increases cumulative interfacial roughness, which degrades the magnetic anisotropy for the top CFB layer.

Temperature-dependent exchange bias measurements (Fig. 3, inset) show the exchange bias vanishing at a blocking temperature (T_B) of ~ 85 K. This drastic reduction from the bulk NiO Néel temperature ($T_N = 523$ K) is a known finite-size effect in thin AFMs, where T_B decreases while effective T_N increases. The Néel temperature and also the existing interlayer effective field on these samples are beyond the scope of this concise report.

B. Isolated Top CFB and Training Effect

Further analysis of the barrier can be done using a sample with the top CFB layer: Buffer/NiO(3)/CFB(4)/Capping. This

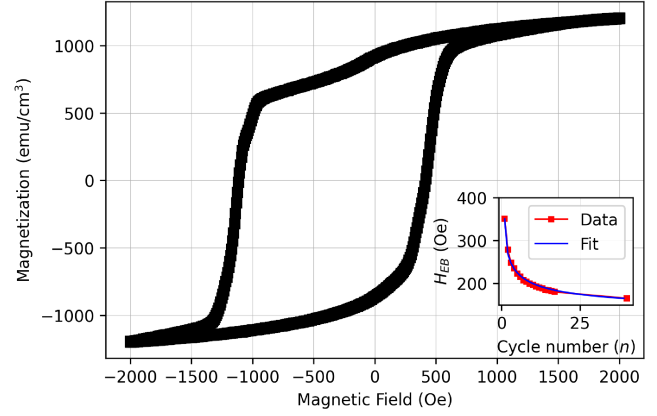


Fig. 4. Hysteresis loop of the control sample with only the top CFB layer. Inset shows exchange bias field versus number of magnetic cycles (training effect).

is also a control sample to confirm the top CFB with lower anisotropy exhibits exchange bias. The hysteresis loop, in Fig. 4, confirms the expected exchange bias behavior. To analyze the exchange interaction on the antiferromagnet-ferromagnet interface, we plotted the exchange bias field against number of magnetic field sweeps (cycle number) (Fig. 4, inset). On the observed training effect, we found the relaxation constant ($\gamma = 7.03 \times 10^{-6} \text{ Oe}^{-2}$) and the residual exchange bias field ($H_\infty = 124$ Oe) an infinite number of field sweeps via the Binek model.

III. CONCLUSION

Magnon-assisted STT switching requires an antiferromagnetic barrier with sizable a TMR and a robust exchange interaction at the interface between the ferromagnetic electrodes and the AFM barrier to ensure efficient magnon transport. We achieved a significantly higher TMR compared to MTJs with only NiO barrier reported in the literature. The antiferromagnetic phase of the NiO barrier was confirmed by the high coercive field and exchange bias observed in as-deposited samples with a T_B of ~ 85 K. Finally, a sizable H_∞ indicates that the exchange interaction at the AFM/FM interface is strong. The short-term goal is to realize energy-efficient magnon-assisted STT switching in nano-pillars.

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